Does Language Matter?
Evaluating LLM-Agent Performance Across Languages

Anonymous ACL submission

Abstract

LLM agents are increasingly deployed to carry out multi-step, tool-using work, yet they are evaluated almost exclusively in English. We ask a simple question with a non-obvious answer: when the *same* task is expressed in a different language, does the agent behave differently, and if so, *where* in its pipeline? We build a paired bilingual (English/Bangla) version of OFFICEBENCH, holding the environment, the tool schema and the checkers fixed while varying only the language of the system prompt and the task instruction, and we run two open-weight models (31B and 20B parameters) over 300 subtasks in each condition—1,200 agent executions in total. We find that the effect of language is **model-dependent in direction**: the 31B model loses 9.7 points in Bangla ($p<0.001$, exact McNemar) while the 20B model *gains* 5.6 points ($p=0.043$). As a consequence, model ranking is language-dependent: a gap that is overwhelming in English ($p=5\times10^{-9}$) becomes statistically indistinguishable in Bangla ($p=0.52$). A failure-mode analysis explains the divergence. The 31B model's failures are *content-bound* and degrade when the instruction language changes; the 20B model's failures are *action-format-bound*—96% of its English failures involve a malformed action—and the language shift reduces the prose leakage that causes them. Finally, we show that a substantial share of the measured Bangla penalty is *evaluator-induced*: 26% of the 31B model's Bangla failures are answers written in Bangla script and rejected by English keyword checkers. Crediting them would raise its Bangla score above its English score. We argue that multilingual agent evaluation requires language-aware verification, and we release code, traces and failure codes.

1 Introduction

Language-model agents now plan over many steps, call tools, and edit real artefacts— spreadsheets, calendars, documents, email (Yao et al., 2023; Liu et al., 2024; Zhou et al., 2024). The benchmarks that measure them, however, are written almost entirely in English (Jimenez et al., 2024; Xie et al., 2024; Mialon et al., 2024). This is a consequential gap. Most of the world does not work in English, and an agent that edits a payroll spreadsheet is supposed to be usable by the person who owns the spreadsheet, in that person's language.

The usual assumption is that moving a task into a lower-resource language degrades performance, because multilingual capability lags English in knowledge and reasoning benchmarks (Ahuja et al., 2023; Shi et al., 2023; Bandarkar et al., 2024; Huang et al., 2023). For *agents* this assumption has not been tested carefully, and it hides a more interesting question. An agentic episode is not one forward pass. It is a loop of reading state, reasoning, emitting a structured action, and consuming the result. Language could in principle perturb any of these stages, and a benchmark score collapses all of them into a single bit of success.

We therefore ask three questions. **RQ1:** Does expressing an equivalent task in a different language change end-to-end agent success, and in which direction? **RQ2:** Where does the change act—on task understanding and content, or on the mechanics of tool interaction? **RQ3:** How much of an apparent cross-lingual gap is a property of the *agent*, and how much is a property of the *evaluation harness*?

To answer these we construct a strictly paired bilingual protocol over OFFICEBENCH (Wang et al., 2025), a benchmark of 300 office subtasks requiring one, two or three applications. We translate only the system prompt and the task instruction into Bangla—a language with roughly 240M speakers that is nevertheless under-represented in LLM training data (Joshi et al., 2020)—while holding the Docker environment, the file state, the tool schema and the automatic checkers byte-

identical across conditions. We evaluate two open-weight models of comparable but distinct scale and lineage, and we code every failure.

Our findings overturn the simple "lower-resource is worse" expectation:

- **Direction is model-dependent.** `gemma4:31b` falls from 44.0% to 34.3% in Bangla ($p<0.001$); `gpt-oss:20b` *rises* from 26.7% to 32.3% ($p=0.043$). Aggregate "multilingual degradation" is not a property of the language.
- **Rankings do not transfer.** The English gap between the two models is enormous ($p=5\times10^{-9}$); in Bangla it disappears ($p=0.52$). A leaderboard built in English does not predict the ordering in another language.
- **Failure mode predicts direction.** The models fail for different reasons. When failures are format-bound, a language shift can help; when they are content-bound, it hurts.
- **Evaluators are part of the measurement.** 26% of the 31B model's Bangla failures are Bangla-script answers rejected by English keyword checks. The measured language penalty is, in large part, a property of the checker.

2 Related Work

Agent benchmarks. Interactive benchmarks evaluate LLMs as decision makers rather than text producers: AGENTBENCH (Liu et al., 2024), WEBARENA (Zhou et al., 2024), OSWORLD (Xie et al., 2024), SWE-BENCH (Jimenez et al., 2024) and GAIA (Mialon et al., 2024). OFFICEBENCH (Wang et al., 2025) is the setting we adopt: it requires switching between office applications and grounding actions in a real file system, and it scores outcomes with execution-based checks rather than string overlap. All of these are English-only.

Multilingual evaluation. Cross-lingual evaluation of LLMs is well developed for static tasks—XNLI (Conneau et al., 2018), MEGA (Ahuja et al., 2023), MGSM (Shi et al., 2023), BELEBELE (Bandarkar et al., 2024)—and consistently reports an English advantage, with the gap widening for lower-resource languages (Huang et al., 2023; Zhang et al., 2023). Etxaniz et al. (2024) show that models often perform better when a prompt is first translated into English, implying an English-centric internal pivot. These studies evaluate single-turn outputs; we ask what happens when the output must be a *valid action in a loop*.

Tool use and structured output. Agents must emit well-formed, schema-conforming calls (Schick et al., 2023; Qin et al., 2024; Patil et al., 2024). Format adherence is itself a capability, and it is known to be brittle. Our analysis shows that this brittleness, rather than comprehension, dominates one of our two models and determines the sign of its language effect.

Failure analysis. Rather than reporting a single score, a growing line of work dissects *how* systems fail (Bang et al., 2023). We follow that approach: we code every failed episode and treat the code distribution, not the aggregate, as the object of study.

3 Study Design

3.1 Benchmark and environment

OFFICEBENCH provides 300 subtasks spanning eight applications (calendar, excel, ocr, pdf, shell, word, email, and an in-container `llm` helper). Tasks are stratified by the number of applications required: 93 one-app, 95 two-app and 112 three-app subtasks. Each subtask ships a seeded file system and a set of execution-based checks (cell values, file existence, calendar overlap, keyword containment). A subtask passes only if *all* of its checks pass. Every episode runs in a fresh Docker container reset from a pinned image.

3.2 The paired bilingual protocol

Our central design requirement is that language be the *only* variable. Concretely, across the two conditions we hold fixed: the container image and seeded files; the available applications; the tool schema (action names and argument keys); the step budget; the decoding parameters; and the checkers. We vary:

1. **The system prompt**, authored natively in Bangla, including the role description, the application descriptions and the response-format instruction.
2. **The task instruction**, machine-translated into Bangla and cached so that every model sees an identical string for a given subtask.

Crucially, the **tool schema remains in English**. Action identifiers such as `switch_app` and argument keys such as `file_path` are API symbols that the executor matches literally; translating

them would change the environment rather than the language of the task, and would make the two conditions incomparable. We therefore present the schema in English in both conditions and attach a Bangla gloss describing what each tool does. This isolates *instruction language* from *interface language*, and it mirrors deployment reality, where an interface is typically English while the user is not.

3.3 Agent scaffold

We use a ReAct-style loop (Yao et al., 2023). At each step the model receives the system prompt, the task, the history of previous (action, observation) pairs, and the actions available in the current application. It must reply with a short reasoning segment followed by exactly one JSON action:

```
##Thought: <reasoning>
##Command: {"app": ...,
"action": ..., ...}
```

The harness parses the JSON after the last `##Command:` marker, so reasoning may not corrupt the action. An episode terminates when the agent emits `finish_task`, repeats an identical action five times (recorded as *stuck*), or reaches the 20-step budget. We log the full trajectory—prompt, reasoning, action, observation—for every step.

3.4 Models

We evaluate two open-weight models served through the same API with identical decoding (temperature $= 0$, top_p $= 1$): `gemma4:31b` (32.7B parameters) and `gpt-oss:20b` (20.9B parameters).[1] The two are deliberately different in kind: the former is a conventional instruction-following model, the latter exposes a separate reasoning channel, and this architectural difference turns out to drive our central result. For the reasoning model we fold the reasoning channel into the logged `##Thought` so that both conditions are logged identically.

3.5 Metrics and statistics

Our primary metric is the benchmark's own pass rate. Because the design is *paired*—the same 300 subtasks in both conditions—we test differences with the exact two-sided McNemar test over discordant pairs, which is the appropriate test for paired binary outcomes (McNemar, 1947). We report discordant counts (b, c) alongside each p-value so that effect size is visible.

[1] Served via Ollama; sizes read from the served model metadata.

Model	Lang	1-app (93)	2-app (95)	3-app (112)	All (300)
gemma4:31b	EN	66.7	54.7	16.1	**44.0**
	BN	45.2	45.3	16.1	34.3
gpt-oss:20b	EN	44.1	34.7	5.4	26.7
	BN	53.8	38.9	8.9	32.3

Table 1: Pass rate (%) by condition and task complexity. Paired exact McNemar: gemma EN vs BN $p = 8.2\times10^{-5}$ (b=41, c=12); gpt-oss EN vs BN $p = 0.043$ (b=23, c=40).

3.6 Failure coding

We code every failed episode from its trajectory. Besides a general taxonomy (benchmark defect, wrong content, out of steps, premature finish, task read differently, action format), the Bangla condition requires codes that cannot occur in English, notably *answer language*: the agent produced a semantically reasonable answer in Bangla script which an English keyword check rejects. In addition to coding, we extract fully objective signals for all 1,200 episodes: whether any action was rejected as malformed, whether the episode ended stuck, whether it exhausted the step budget, whether it finished and still failed, episode length, and whether the final answer contained Bangla script.

4 Results

4.1 RQ1: Language changes success, but not in one direction

Table 1 gives the main result. The two models move in *opposite* directions. `gemma4:31b` drops from 44.0% to 34.3%, a 9.7-point loss, with 41 subtasks passing only in English against 12 passing only in Bangla (exact McNemar $p = 8.2\times10^{-5}$). `gpt-oss:20b` moves the other way, from 26.7% to 32.3%, a 5.6-point gain, with 23 English-only passes against 40 Bangla-only passes ($p = 0.043$).

The aggregate conclusion is therefore not "Bangla is harder". The language manipulation is identical; only the model differs. Any claim of the form "model performance degrades in language L" is under-specified unless the model is named.

Rankings do not transfer across languages. This has a direct practical consequence (Table 2).

Comparison	b	c	exact p
gemma vs. gpt-oss (EN)	67	15	5.3×10^{-9}
gemma vs. gpt-oss (BN)	33	27	0.52

Table 2: The model gap is overwhelming in English and vanishes in Bangla. b: solved only by gemma; c: solved only by gpt-oss.

In English, `gemma4:31b` beats `gpt-oss:20b` decisively: 67 subtasks are solved only by gemma against 15 only by gpt-oss ($p = 5.3 \times 10^{-9}$). In Bangla the same comparison yields 33 against 27— statistically indistinguishable ($p = 0.52$). A practitioner choosing a model from an English leaderboard would believe the first model is far better; in Bangla that advantage is not measurable.

4.2 RQ2: Complexity gates the language effect

Table 1 also shows that the language effect is not uniform across difficulty. For `gemma4:31b` the Bangla penalty is largest on one-app tasks (-21.5 points) and *exactly zero* on three-app tasks (16.1% in both conditions).

The explanation is a floor effect. Three-app subtasks require long multi-application workflows—read an image, OCR it, summarise, split the result, send mail—and both models fail the large majority of them regardless of language. Where a model has no competence to lose, language cannot take any away. The corollary matters for benchmark design: a cross-lingual evaluation run only on hard tasks will under-report the language effect, because the hardest items are least sensitive to it.

4.3 RQ3: Failure mode determines the sign

Why do the two models move in opposite directions under an identical manipulation? Because they fail for different reasons (Table 3).

`gpt-oss:20b` is **format-bound**. 212 of its 220 English failures (96%) contain at least one action rejected as malformed. Its reasoning lives in a separate channel, and in English it frequently spills prose into the action slot, producing a rejected action, then another, until the episode is consumed. Under Bangla instructions this leakage falls: malformed-action failures drop to 186, stuck episodes halve (30 → 14), budget exhaustion falls (81 → 58), and episodes shorten (13.3 → 12.1 steps). The model is not understanding the task better; it is *talking less* into the action channel, and it therefore completes more episodes. Its gain in Bangla is a format effect, not a comprehension effect.

	gemma4:31b		gpt-oss:20b	
Signal	EN	BN	EN	BN
Failures	168	197	220	203
malformed action	53	23	**212**	**186**
ended stuck	3	5	30	14
out of steps	46	39	81	58
finished, failed	118	152	108	130
Mean steps/episode	10.9	10.5	13.3	12.1
Bangla-script answer	0	**51**	0	7

Table 3: Failure composition over all 300 subtasks per condition. The three terminal states (stuck / out of steps / finished-but-failed) are mutually exclusive and sum to the failure count together with one unsaved episode per condition; *malformed action* is a cross-cutting flag counting failures in which at least one action was rejected, so it overlaps the others. gpt-oss is dominated by action-format failures; gemma by episodes that finish and still fail.

`gemma4:31b` is **content-bound**. It emits valid actions (only 53 of 168 English failures involve a malformed action) and its dominant failure is an episode that runs to completion and is then rejected on content—118 in English rising to 152 in Bangla. This is the mode that a language change should affect, and it does.

This yields a simple predictive rule, and the central claim of this paper:

> *The sign of an agent's cross-lingual effect follows its dominant failure mode. Content-bound agents lose under a language shift; format-bound agents may gain, because the shift suppresses the verbosity that corrupts their actions.*

4.4 RQ4: Much of the measured penalty belongs to the evaluator

The last row of Table 3 isolates a failure mode that *cannot* occur in English. In 51 of its 197 Bangla failures (26%), `gemma4:31b` terminated by calling `finish_task` with an answer written in Bangla script. For 49 of those subtasks the failing check is `evaluate_contain`, which tests whether an English keyword occurs in the output.

In other words: the agent was asked a question in Bangla, answered in Bangla, and was marked wrong because the checker was written in English. This is not an agent capability failure. It is a property of the measuring instrument.

Condition	Strict	Lenient bound	EN reference
gemma4:31b BN	34.3	≤ 51.3	44.0
gpt-oss:20b BN	32.3	≤ 34.7	26.7

Table 4: Sensitivity of the Bangla score to evaluator language (%). The lenient bound credits every Bangla-script answer. For gemma the bound brackets—and exceeds—the English score.

Table 4 reports the sensitivity. Under the strict (as-shipped) checkers `gemma4:31b` scores 34.3% in Bangla. If every Bangla-script answer were credited—a deliberate upper bound, since we do not verify each one semantically—it would score 51.3%, which is *above* its own English score of 44.0%. The true value lies between the two bounds, and we cannot determine it with English-only checkers. The honest conclusion is therefore stronger and more useful than the naive one: **we cannot claim from these numbers that Bangla degrades this model's task competence, because the instrument that produced the number is itself language-biased.**

Notably the confound is itself model-dependent: `gpt-oss:20b` answered in Bangla only 7 times, because it remains English-anchored even under Bangla instructions. A checker bias thus penalises precisely the model that adapts to the user's language—exactly the behaviour one would want to reward.

5 Discussion

"Multilingual degradation" is not one phenomenon. Our two models received an identical manipulation and moved in opposite directions, with the sign predicted by their failure composition. Reporting a single aggregate delta per language hides this. We recommend that cross-lingual agent evaluations report failure-mode composition alongside the score; it is both more informative and, as we show, necessary to interpret the sign.

Leaderboards are language-scoped artefacts. The ranking between our two models is decisive in English and absent in Bangla. Deployment decisions for non-English users should not be read off English leaderboards.

Evaluation harnesses need language-aware verification. Execution-based checks are usually presented as more robust than surface matching. They are—for file state and cell values. But a large fraction of office tasks are questions whose answer is checked by keyword, and those checks silently encode a language. Multilingual agent benchmarks should either verify answers semantically, accept the task language, or separate "artefact-state" checks from "answer-text" checks and report them apart. Our Table 4 is the minimum diagnostic we would want to see reported.

Interface language versus instruction language. Keeping the tool schema in English was a methodological necessity, but it reflects a real deployment regime and it isolates a useful finding: an agent can be instructed in one language while acting through an English-symbol API, and the main cost is not symbol confusion but answer-language mismatch at the boundary where the agent reports back to the user.

6 Conclusion

We built a strictly paired English/Bangla agent evaluation over OFFICEBENCH and ran 1,200 episodes across two open-weight models. Task language changes agent success, but the direction depends on the model's dominant failure mode: a content-bound model loses 9.7 points, while a format-bound model gains 5.6. Model rankings that are decisive in English can vanish in another language. And a quarter of one model's Bangla failures are artefacts of English-only checkers, enough to bracket its true Bangla score above its English score. Language matters for LLM agents—but how it matters is a property of the model and of the evaluator, not of the language alone.

Limitations

One non-English language. We study Bangla only. Bangla is a useful probe—widely spoken yet under-represented—but our directional claim (sign follows failure mode) should be tested on typologically different and higher-resource languages before being treated as general.

Machine-translated instructions. Task instructions were machine-translated and cached; the system prompt was authored natively. Translation noise is therefore part of the Bangla condition, and some failures we code as content errors may be translation artefacts. We mitigated this by using one fixed translation per subtask for both models,

so the comparison *between* models is unaffected, but the absolute Bangla scores should be read as a lower bound on achievable performance.

Two models. Two models cannot establish a general law. They were chosen to differ in kind (conventional versus separate-reasoning-channel), which is what exposes the mechanism, but the mechanism now needs replication across more systems. An evaluation of a third model in the same size class was in progress at submission time and is not reported here.

Decoding is not perfectly reproducible. We decode at temperature 0, but served inference is not bit-deterministic: batching and floating-point non-associativity can change an argmax when two tokens are near-tied, and a single divergence early in an episode can flip the outcome. We did not re-run conditions to estimate this variance directly, so part of the smaller effect we report (the 5.6-point gpt-oss gain, $p = 0.043$) may be run-to-run noise. The larger gemma effect ($p < 0.001$) and the model-gap collapse are well outside plausible noise, but a replication study would strengthen the weaker claim. We regard this as the most important missing control.

The lenient bound is an upper bound. We did not semantically verify each of the 51 Bangla-script answers; the bound credits them all. The true Bangla score lies strictly between the two bounds and we do not claim to have located it.

Infrastructure effects. Episodes ran on a memory-constrained host. A small number of episodes (at most one per condition in the reported runs) failed to persist their output and are counted as failures, and a per-episode wall-clock cap occasionally terminated a slow episode. These affect all conditions approximately equally and are too few to alter any reported comparison, but they are not zero.

Ethics Statement

This work evaluates publicly available open-weight models on a public benchmark in a sandboxed container; no human subjects and no private data are involved. Our main finding has a direct equity dimension: evaluation harnesses written in English can penalise an agent for answering in the user's language, which systematically understates agent quality for non-English users and rewards models that ignore the user's language. We release our bilingual protocol, trajectories and failure codes so that this bias can be measured rather than inherited. The translated task set was produced by machine translation and may contain errors; it is intended for evaluation research, not as a curated Bangla language resource.

References

Kabir Ahuja, Harshita Diddee, Rishav Hada, Millicent Ochieng, Krithika Ramesh, Prachi Jain, Akshay Nambi, Tanuja Ganu, Sameer Segal, Maxamed Axmed, Kalika Bali, and Sunayana Sitaram. 2023. MEGA: Multilingual evaluation of generative AI. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Lucas Bandarkar, Davis Liang, Benjamin Muller, Mikel Artetxe, Satya Narayan Shukla, Donald Husa, Naman Goyal, Abhinandan Krishnan, Luke Zettlemoyer, and Madian Khabsa. 2024. The Belebele benchmark: a parallel reading comprehension dataset in 122 language variants. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*.

Yejin Bang, Samuel Cahyawijaya, Nayeon Lee, Wenliang Dai, Dan Su, Bryan Wilie, Holy Lovenia, Ziwei Ji, Tiezheng Yu, Willy Chung, Quyet V. Do, Yan Xu, and Pascale Fung. 2023. A multitask, multilingual, multimodal evaluation of ChatGPT on reasoning, hallucination, and interactivity. In *Proceedings of the 13th International Joint Conference on Natural Language Processing (IJCNLP-AACL)*.

Alexis Conneau, Ruty Rinott, Guillaume Lample, Adina Williams, Samuel R. Bowman, Holger Schwenk, and Veselin Stoyanov. 2018. XNLI: Evaluating cross-lingual sentence representations. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Julen Etxaniz, Gorka Azkune, Aitor Soroa, Oier Lopez de Lacalle, and Mikel Artetxe. 2024. Do multilingual language models think better in English? In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*.

Haoyang Huang, Tianyi Tang, Dongdong Zhang, Wayne Xin Zhao, Ting Song, Yan Xia, and Furu Wei. 2023. Not all languages are created equal in LLMs: Improving multilingual capability by cross-lingual-thought prompting. In *Findings of the Association for Computational Linguistics: EMNLP 2023*.

Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. 2024. SWE-bench: Can language models resolve real-world GitHub issues? In *International Conference on Learning Representations (ICLR)*.

Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. 2020. The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*.

Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, and 3 others. 2024. AgentBench: Evaluating LLMs as agents. In *International Conference on Learning Representations (ICLR)*.

Quinn McNemar. 1947. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157.

Grégoire Mialon, Clémentine Fourrier, Craig Swift, Thomas Wolf, Yann LeCun, and Thomas Scialom. 2024. GAIA: A benchmark for general AI assistants. In *International Conference on Learning Representations (ICLR)*.

Shishir G. Patil, Tianjun Zhang, Xin Wang, and Joseph E. Gonzalez. 2024. Gorilla: Large language model connected with massive APIs. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, Sihan Zhao, Lauren Hong, Runchu Tian, Ruobing Xie, Jie Zhou, Mark Gerstein, Dahai Li, Zhiyuan Liu, and Maosong Sun. 2024. ToolLLM: Facilitating large language models to master 16000+ real-world APIs. In *International Conference on Learning Representations (ICLR)*.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Freda Shi, Mirac Suzgun, Markus Freitag, Xuezhi Wang, Suraj Srivats, Soroush Vosoughi, Hyung Won Chung, Yi Tay, Sebastian Ruder, Denny Zhou, Dipanjan Das, and Jason Wei. 2023. Language models are multilingual chain-of-thought reasoners. In *International Conference on Learning Representations (ICLR)*.

Zilong Wang, Yuedong Cui, Li Zhong, Zimin Zhang, Da Yin, Bill Yuchen Lin, and Jingbo Shang. 2025. OfficeBench: Benchmarking language agents across multiple applications for office automation. In *Findings of the Association for Computational Linguistics: NAACL 2025*. ArXiv:2407.19056.

Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Lu, Kai Tan, Yanbo Shen, Zhennan Liu, Yuchen Guo, Zhiwen Wang, Yiheng Qi, et al. 2024. OSWorld: Benchmarking multimodal agents for open-ended tasks in real computer environments. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*.

Xiang Zhang, Senyu Li, Bradley Hauer, Ning Shi, and Grzegorz Kondrak. 2023. Don't trust ChatGPT when your question is not in english: A study of multilingual abilities and types of LLMs. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. 2024. WebArena: A realistic web environment for building autonomous agents. In *International Conference on Learning Representations (ICLR)*.

A Reproducibility details

Each episode runs in a fresh container reset from a pinned image, with a 20-step budget and a single seeded file system per subtask. Decoding is fixed at $\mathrm{temperature} = 0$, $\mathrm{top_p} = 1$. The harness records, per step, the full prompt, the model reply, the parsed action and the environment observation; all analyses in this paper are computed from those logs and from the benchmark's own scorer output, not from intermediate summaries.

B Failure code book

Code	Meaning
BM	Benchmark defect: a check or reference asks for something the task does not state
WC	Wrong content: output exists, a value is wrong
SB	Out of steps: hit the 20-step budget
PF	Premature finish: declared done with a required output missing
TR	Task read differently: a different operation from the one asked
FMT	Action format: could not produce a valid action
ER	Empty result taken as "nothing exists"
Bangla-condition only	
AL	Answer language: correct answer in Bangla script, English check fails
NC	Name case: a name rewritten from Bangla script with wrong case
WD	Word choice when translating a literal back to English
TX	Translation changed the task or a literal

Table 5: Failure codes. AL–TX cannot occur in the English condition and are the language-specific modes discussed in Section 4.4.